KK-W-231D
September 19, 1986
SUPERSEDING
KK-W-231C
February 12, 1969

FEDERAL SPECIFICATION

WELTING, LEATHER, SHOE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one grade of vegetable tanned leather welting.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications

A-A-203 - Paper, Kraft, Untreated
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8330

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Federal Standards

FED-STD-123 - Marking for Shipment (Civil Agencies)
FED-STD-311 - Leather, Methods of Sampling and Testing

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Material. The leather used for the welting shall be full grain vegetable tanned cattlehide in the form of double shoulders. The leather shall be lightly buffed on the grain side.

3.3 Construction.

3.3.1 Fillet strips. The welting shall be made up of fillet leather strips. The strips shall be joined end to end to form a hank of welting. No strip in a hank shall be less than 7 inches in length, and not more than 7 percent of the strips in a hank shall be less than 22 inches in length. Strips having lengths of 22 inches or less shall not occur consecutively in a hank.

3.3.2 Skiving. Both ends of each fillet strip shall be skived to a feather edge.

3.3.3 Laps. The skived flesh end and the skived grain end of contiguous strips shall be joined with waterproof cement to form a lap of not less than 1/2 inch or not more than 1 1/2 inches in length. The ends of the laps shall be approximately at right angles to the edges of the strips. All edges of the joint shall be flush.

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3.4 Physical requirements.

3.4.1 Groove and bevel. Unless otherwise specified (see 6.2), the welting shall be beveled and grooved conforming to the dimensions of table I (see figure 1).

TABLE I. Welting dimensions (inches)

Welting width	Welting thick-ness	Groove width	Groove depth	Edge to groove center	Bevel width	Bevel thickness at butt	Tolerances all dimen-sions
1/2	1/8	5/64	0.050	9/64	0.100	0.050	±1/128 (0.0078)

3.4.2 Finish. The welting shall be lightly buffed on the grain side. Unless otherwise specified (see 6.2), a finish containing dye and pigments only in amounts sufficient to provide a uniform color shall be applied to the grain side.

3.4.3 Color. Unless otherwise specified, the color of the welting shall be black (see 6.2).

3.4.4 Cracking. The welting shall not crack when tested as specified in 4.4.5.

3.4.5 Length of hank. The length of hank shall be 100 yards (-0, +1 foot).

3.5 Chemical requirements.

3.5.1 Welting leather. The welting leather shall conform to the chemical requirements of table II when tested as specified in 4.4.5.

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TABLE II. Chemical requirements

Characteristic	Minimum (percent)	Maximum (percent)
Chloroform-soluble material 1/	8.0	-
Acidity (pH)	3.0	-
Paranitrophenol fungicide 2/	0.18	0.70

1/ Moisture free basis.

2/ Unless otherwise specified in 6.2.

3.6 Workmanship. The finished leather welting shall conform to the quality of product established by this specification, and the occurrence of defects shall not exceed the applicable acceptable quality levels (AQL).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with FED-STD-311.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table III. The lot size shall be expressed in units of hanks of welting. The sample unit shall be one hank of welting. The sample size and acceptance/rejection criteria shall be in accordance with FED-STD-311.

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TABLE III. End item visual defects

Examine	Defect
Finish	Unless otherwise specified, grain side not dyed and finished Finished grain chipped, peeled, cracked, abraded, or scratched
Color	Unless otherwise specified, color not black Color of grain side not uniform
Quality of leather	Cut, hole, or tear Loose, spongy, pipey, or stretchy area Brittle, boney, or boardy area Flesh side not free of loose flesh Stain or foreign matter Deep wrinkles or any surface defects affecting appearance
Construction and workmanship	Bevel or groove missing or not as required Laps not completely bonded Lap ends not at approximate right angles to edges of the strips Edges of lap joint not flush Bevel or groove not uniformly accomplished

4.4.4 End item dimensional examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of hanks of welting. The sample unit shall be one hank of welting. The sample size and acceptance/rejection criteria shall be in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for dimensional examinations in lots consisting of 91 or more hanks.

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TABLE IV. End item dimensional defects

Examine	Defect
All dimensions (except length of fillet strip) 1/	Any dimension not as specified in table I
Length of fillet strip 2/	Any strip less than 7 inches in length More than 7 percent of strips less than 22 inches in length Strips less than 22 inches in length occur consecutively

1/ Five determinations shall be made at random on each sample unit.

2/ All fillet strips in each sample unit shall be examined for length and shall be measured from the point of the lap on one end of the strip to the heel of the other end of the strip. Sampling procedure for this examination shall be in accordance with MIL-STD-105.

4.4.5 End item testing. The end item shall be tested in accordance with the methods specified in FED-STD-311 wherever applicable and as listed in table V. The sample unit shall be one 12-inch length of welting that is free of laps. The sample size shall be 15. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be rejected if any of the following conditions exist:

a. Any test failure occurs for cracking.
b. Any composite test result fails to meet the specified requirement.

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TABLE V. End item tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite sample
Material identification	3.2	1/	-	-
Waterproof cement	3.3.3	1/	-	-
Cracking	3.4.4	4021 2/	X	-
Chloroform-soluble material	3.5.1	6341	-	X
Acidity (pH)	3.5.1	6621	-	X
Paranitrophenol	3.5.1	6711	-	X

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the requirement stated.

2/ Except that the specimen shall be a 15-cm length of welting. The test procedure shall be repeated twice, once by bending the specimen with the grain side out and again by bending the specimen with the grain side in. The diameter of the mandrel shall be 1/4 inch.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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Examine	Defect
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

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5.1.1 Level A. Each hank of welting shall be secured with cotton tape or twine approximately 4 inches in from each end. Each hank shall then be completely wrapped in a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. The wrap shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial. Welting shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Five hanks of welting, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Five hanks of welting, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.1 Weather-resistant shipping containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Welting, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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5.3 Palletization (Military requirements). When specified (see 6.2), hanks of welting, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. The pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with FED-STD-123 or ASTM D 3951, as applicable.

5.4.2 Military agencies. In addition to any special marking required by the contract or purchase order; unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The leather welting is intended for use in footwear.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. When groove and bevel are not required; dimensions when not as specified (see 3.4.1).
d. When dye and finish is not required (see 3.4.2).
e. Color, when black color is not required (see 3.4.3).
f. When paranitrophenol is not required (see 3.5.1).
g. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
h. Type and class of unit load required (see 5.2.1).
i. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Civil agency procurements. When level B packaging requirements are specified for Civil agency procurements, the requirements in 5.1.1 shall apply.

6.5 Subject term (key word) listing.

Footwear
Welting, leather shoe

MILITARY INTERESTS:

Custodians

Army - GL
Navy - NU
Air Force - 99

Review activities

Army - MD
Air Force - 82, 11
DLA - CT

User activity

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army - GL

Project No. 8330-0109

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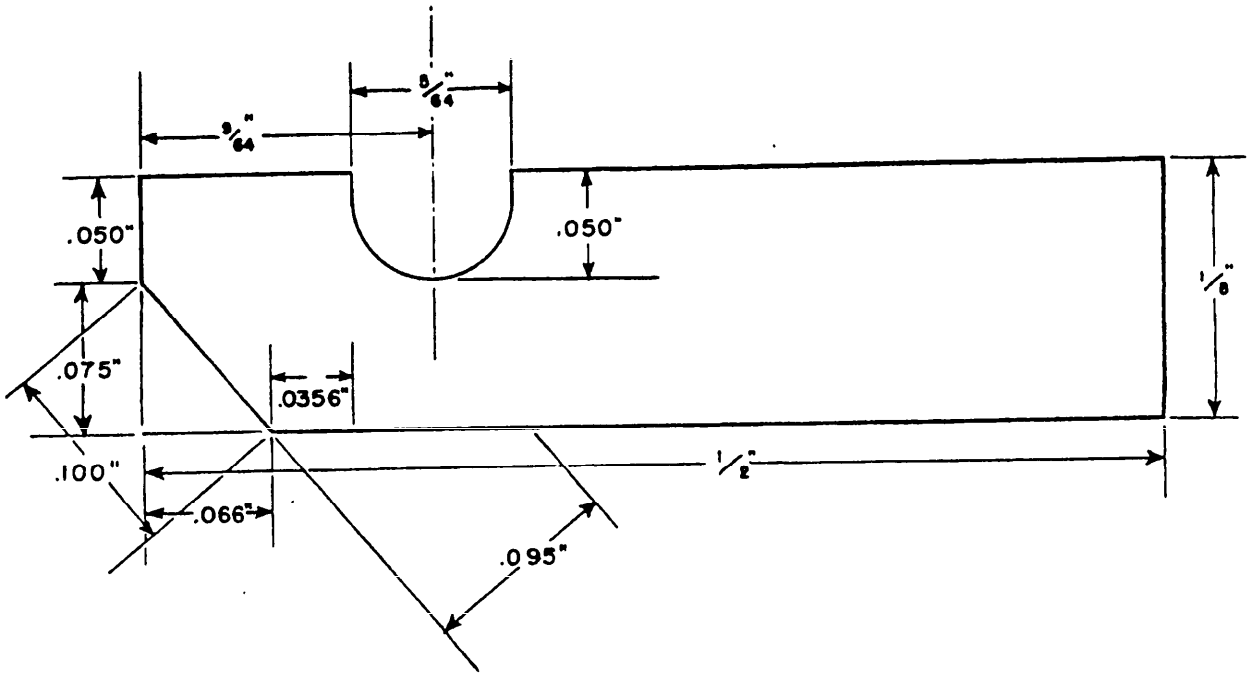


Figure 1. Welting, Leather

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
KK-W-231D	Welting, Leather, Shoe

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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